FOLD Progress

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May 12, 2025

New Metric: Coherence

- Cross-Correlation measures similarity of X and Y through time shifts
- Fourier Transform of Cross-Correlation between X and Y is Cross-Spectral Density
- Coherence is normalized Cross-Spectral Density, how strongly two signals are linearly related at each frequency

Coherence Trends

- For likelihood, acheives significant linear correlation with mutual intelligibility
- For both likelihood and embeddings, much less correlated with FSI

Likelihood `compute_overlaps` (600, 35, 35) (previous results)

metric	p_coef	p_pval	s_coef	s_pval	num_points
fsi	-0.143	0.419	-0.607	0.000	34
lex_sim	-0.801	0.000	-0.861	0.000	56
mut_int	-0.017	0.902	0.049	0.728	52
pho_sim	-0.656	0.000	-0.473	0.000	496

Likelihood `kl_divergence_matrix` (previous results)

metric	p_coef	p_pval	s_coef	s_pval	num_points
fsi	0.111	0.532	0.677	0.000	34
lex_sim	0.800	0.000	0.830	0.000	56
mut_int	-0.010	0.944	-0.114	0.422	52
pho_sim	0.407	0.000	0.486	0.000	496

Likelihood `coherence_fun`

metric	p_coef	p_pval	s_coef	s_pval	num_points
fsi	-0.026	0.884	-0.178	0.313	34
lex_sim	-0.795	0.000	-0.849	0.000	56
mut_int	0.405	0.003	0.485	0.000	52
pho_sim	-0.868	0.000	-0.377	0.000	496

Embeddings `compute_overlaps` (previous results)

metric	p_coef	p_pval	s_coef	s_pval	num_points
fsi	-0.078	0.663	-0.539	0.001	34
lex_sim	-0.799	0.000	-0.921	0.000	56
mut_int	0.491	0.000	0.515	0.000	52
pho_sim	-0.855	0.000	-0.681	0.000	496

Embeddings `kl_divergence_matrix` (previous results)

metric	p_coef	p_pval	s_coef	s_pval	num_points
fsi	0.176	0.319	0.510	0.002	34
lex_sim	0.801	0.000	0.899	0.000	56
mut_int	-0.524	0.000	-0.533	0.000	52
pho_sim	0.876	0.000	0.652	0.000	496

Embeddings `coherence_fun`

metric	p_coef	p_pval	s_coef	s_pval	num_points
fsi	0.047	0.791	0.146	0.410	34
lex_sim	0.819	0.000	0.835	0.000	56
mut_int	-0.435	0.001	-0.460	0.001	52
pho_sim	0.850	0.000	0.355	0.000	496

WALS Analysis

26A: Prefixing vs. Suffixing in Inflectional Morphology

- Includes many kinds of affixing, like case affixes on nouns
- Croatian (Strongly suffixing, Postpositional clitics): Dobar [sam]
- English (Strongly Suffixing): Strength -> strength[s], strength[en]

WALS Values:

- value 1 (Little or no inflectional morphology): ['th', 'vi']
- value 2 (Predominantly suffixing): ['bg', 'de', 'en', 'es', 'fi', 'hi', 'hu', 'id', 'it', 'ja', 'ko', 'lt', 'ne', 'no', 'pl', 'pt', 'ru', 'sv', 'tr']
- value 3 (Moderate preference for suffixing): ['am', 'he']
- value 4 (Approximately equal amounts of suffixing and prefixing): ['ar']

Large difference between value 3 (Moderate preference for suffixing) and other groups

78A: Coding of Evidentiality

marks the source of information the speaker has for his or her statement

German (modal morpheme): Er soll krank sein/He is reportedly sick
sollen is a modal verb meaning reportedly

WALS Values:

- value 1 (No grammatical evidentials): ['en', 'es', 'he', 'hi', 'hu', 'id', 'ru', 'th', 'vi']
- value 2 (Verbal affix or clitic) : ['ja', 'ko', 'lt', 'ne']
- value 3 (Part of the tense system): ['bg', 'tr']
- value 5 (Modal morpheme): ['fi']
- value 6 (Mixed systems): ['sv']

No particular differences between groups

Frequency Bands

- Masked all frequencies except those in band, only tested with KL divergences
- Marked increase in Pearson p-value significance around:
  44% - 48%
  60% - 64%
  72% - 76%
- Pearson coef had little change, and Spearman p-value and coef were unaffected

Next Steps

- Frequency band exploration with coherence (probably similar spikes as with kl divergence)
- Replace parts of speech with placeholder, analyze how the spectra changes